Lesson notes · for you

Power from batteries and the sun

IoT · Level 3 · Lesson 4

Materials
Two or three batteries or power banks with mAh on the label; Paper, a pen and a calculator (a phone works); Optional: a small solar garden light, to see a panel and a battery at work; Optional: a phone for the Battery vs how often to send lab

Learning goals

- Explain why a device's radio uses more battery than anything else.
- Make a battery last longer by sleeping and sending less often.
- Describe how a small solar panel keeps a sensor running.

Activity: Battery sums

1. Look on power banks, phone battery boxes and rechargeable batteries for a number followed by mAh. If a label shows none, use about 2,500 mAh for two AA batteries.
2. Multiply each capacity by 0.7. Real batteries give less than the label in heat, in cold and as they age.
3. Divide by the current: hours = mAh ÷ mA. Do it for a device drawing 100 mA, then one drawing 10 mA, and turn big answers into days.
4. Decide what your garden sensor measures, how often it really needs to send, and which radio suits the distance: Bluetooth LE to a phone, Wi-Fi to the school router, or LoRa for a far field.
5. If it sends often or the batteries won't last a term, add a small solar panel, and size the battery for three to five cloudy days.

Never open, crush, puncture or heat a battery, and never leave lithium batteries or power banks in hot sun. Keep button batteries away from small children.

Discussion

- Which data around you must be fresh every second, and which could wait an hour?
- Where at your school would a solar-powered sensor be useful?
- Why does a coin cell suit Bluetooth LE but not Wi-Fi?

Check yourself

- A sensor sends every minute over Wi-Fi. Name two ways to make its battery last longer.

Try it this week
Battery sums: Find two or three batteries or power banks with mAh on the label. For each one, work out how many hours it could run a device drawing 100 mA, and one drawing 10 mA, using 70% of the label. Then plan a sensor for the school garden: what it measures, how often it sends, which radio it uses, how long its batteries would last, and whether it needs a small solar panel. Never open, crush or heat a battery.

Worksheet

Power from batteries and the sun

IoT · Level 3 · Lesson 4

Name _____ Class _____ Date _____

1 **A solar-powered rain sensor works all day but stops every night. What is most likely wrong?**

- ◯ The sensor uses more power at night
- ◯ The panel is broken
- ◯ The battery is too small or worn out

2 **A sensor draws an average of 0.5 mA from two AA batteries (2,500 mAh).**

On paper the batteries last 2,500 ÷ 0.5 = _____ hours, which is about _____ days.

Word bank: 1250 · 52 · 5000 · 208

3 **If a Wi-Fi sensor sends every two minutes instead of every minute, its batteries last roughly twice as long.**

- ◯ True
- ◯ False

4 **Match each radio to what it's good for.**

Wi-Fi	• • • •	Tiny power, but only reaches a phone nearby
LoRa	• • • •	Works anywhere with coverage, but needs a SIM and a plan
Bluetooth LE	• • • •	Fast, but power-hungry; needs a router nearby
Mobile network (4G)	• • • •	Small messages, kilometres away, very little power

5 **A Wi-Fi send takes about 120 mA for 2 seconds; a LoRa send about 40 mA for 0.4 seconds. About how many LoRa sends use the same charge as one Wi-Fi send?**

- ◯ About 3
- ◯ About 5
- ◯ About 15

✎ **Think of a battery-powered device you use every day. What drains its battery most, and what could make it last longer?**

_____

_____

_____

Check your answers

Power from batteries and the sun

IoT · Level 3 · Lesson 4

1 **The battery is too small or worn out**

A broken panel couldn't run the sensor all day, so the panel works.

2 **On paper the batteries last 2,500 ÷ 0.5 = 5000 hours, which is about 208 days.**

Hours = capacity ÷ average current = 2,500 ÷ 0.5 = 5,000 hours, and 5,000 ÷ 24 ≈ 208 days, about seven months.

3 **True**

Sending is almost all of the average current, so halving the sends nearly halves it.

4 **Wi-Fi — Fast, but power-hungry; needs a router nearby**
LoRa — Small messages, kilometres away, very little power
Bluetooth LE — Tiny power, but only reaches a phone nearby
Mobile network (4G) — Works anywhere with coverage, but needs a SIM and a plan

No radio is best at everything. A far field suits LoRa through a gateway, a sensor beside your phone suits Bluetooth LE, a classroom with a router can use Wi-Fi if there's power, and mobile works anywhere with coverage, at a monthly cost.

5 **About 15**

The charge a send uses is current × time.